

Optimal Control

LQR Optimal Control Task 2M

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Var 6

N°	A	b	Q	r
1	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	1
2	$\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	2
3	$\begin{bmatrix} 0 & 1 \\ 4 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	3
4	$\begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	4
5	$\begin{bmatrix} 0 & 1 \\ -2 & 5 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	5
6	$\begin{bmatrix} 0 & 1 \\ 9 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	1

System Description

$$A = \begin{bmatrix} 0 & 1 \\ 9 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$R = 1$$

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Optimal Control Design

The performance index is defined as:

$$J = \int_0^{10} (x^T Q x + u^T R u) dt$$

The optimal control law is given by:

$$u(t) = -K(t)x(t)$$

where the feedback gain matrix is obtained from:

$$K(t) = R^{-1} B^T P(t)$$

The Riccati Differential Equation is:

$$\dot{P}(t) = -A^T P - P A + P B R^{-1} B^T P - Q$$

with terminal condition:

$$P(t) = 0$$

MATLAB Implementation

MATLAB was used to:

- solve the Riccati differential equation,
- compute the optimal feedback gain $K(t)$
- simulate the closed-loop system
- evaluate the cost function
- generate state and control response plots

```
clc;
clear;
close all;

%% System matrices
A = [0 1;
     9 -1];

B = [1;
     0];

Q = [1 0;
     0 2];
```

```

R = 1;

T = 10;

%% Terminal condition
P_T = [0 0;
        0 0];

%% Convert matrix P into vector form
P0 = P_T(:);

%% Riccati Differential Equation
riccati = @(t,P) reshape( ...
    -(A'*reshape(P,2,2) ...
    + reshape(P,2,2)*A ...
    - reshape(P,2,2)*B*(1/R)*B'*reshape(P,2,2) ...
    + Q), ...
    [],1);

%% Solve backward in time
[t,P_sol] = ode45(riccati,[T 0],P0);

%% Reverse time order
t = flipud(t);
P_sol = flipud(P_sol);
%% Compute K(t)
K = zeros(length(t),2);

for i = 1:length(t)

    P = reshape(P_sol(i,:),2,2);

    K(i,:) = (1/R)*B'*P;

end
%% Compute K(t) LQR used in simulink
A = [0 1;
      9 -1];

B = [1;
      0];

Q = [1 0;
      0 2];

R = 1;

K1 = lqr(A,B,Q,R)

```

```
K1 = 1×2  
    6.1583    2.0514
```

```
%% Initial condition  
x0 = [1;  
      0];  
  
%% State storage  
x = zeros(length(t),2);  
  
x(1,:) = x0';  
  
%% Forward simulation  
for i = 1:length(t)-1  
  
    dt = t(i+1)-t(i);  
  
    Kt = K(i,:);  
  
    Acl = A - B*Kt;  
  
    xdot = Acl*x(i,:);  
  
    x(i+1,:) = x(i,:) + dt*xdot';  
  
end  
%% Control input  
u = zeros(length(t),1);  
  
for i = 1:length(t)  
  
    u(i) = -K(i,:)*x(i,:);  
  
end  
  
%% Cost function  
cost = zeros(length(t),1);  
  
for i = 1:length(t)  
  
    xi = x(i,:);  
  
    cost(i) = xi'*Q*xi + R*u(i)^2;  
  
end  
  
J = trapz(t,cost);  
  
disp('Cost function J = ')
```

Cost function $J =$

```
disp(J)
```

5.8060

```
% Plot states
figure;

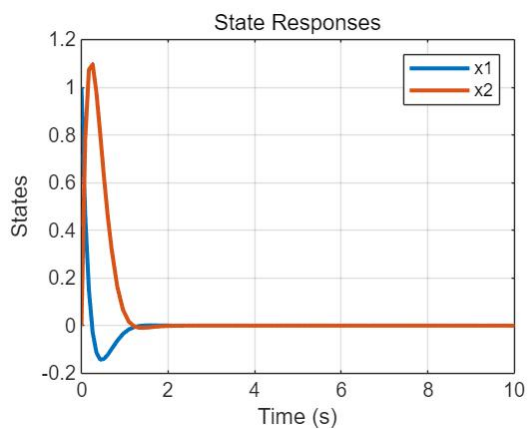
plot(t,x(:,1),'LineWidth',2);
hold on;
plot(t,x(:,2),'LineWidth',2);

xlabel('Time (s)');
ylabel('States');

legend('x1','x2');

title('State Responses');

grid on;
```



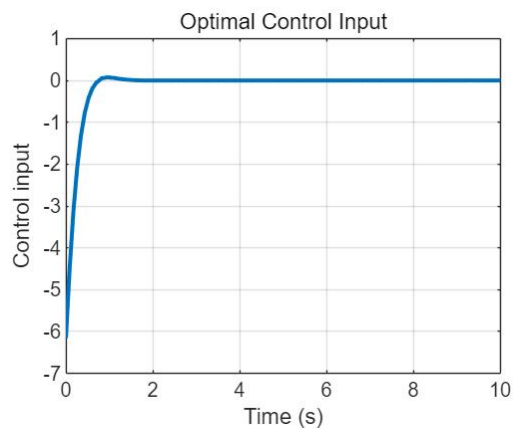
```
% Plot control
figure;

plot(t,u,'LineWidth',2);

xlabel('Time (s)');
ylabel('Control input');

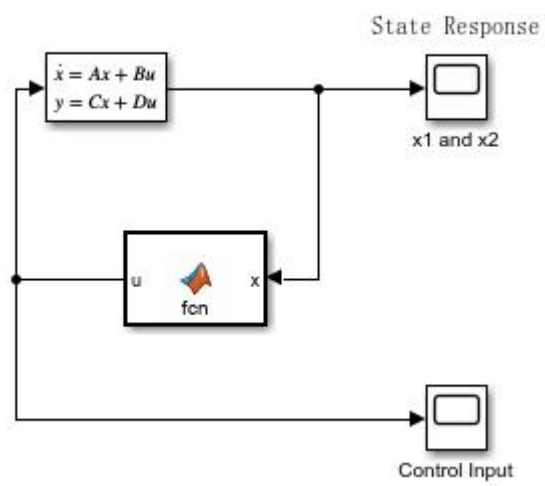
title('Optimal Control Input');

grid on;
```



Solver ode45

Simulink Implementation:



参数

A:

A [0, 1; 9, -1] ⋮

B:

B [1; 0] ⋮

C:

eye(2) [1, 0; 0, 1] ⋮

D:

[0; 0] ⋮

初始条件:

[1; 0] ⋮

参数可调性: 自动 ⌵

☐ 允许最初指定为零的 D 矩阵具有非零值

绝对容差:

auto ⋮

状态名称: (例如, 'position')

''

确定 (O) 取消 (C) 帮助 (H) 应用 (A)

```

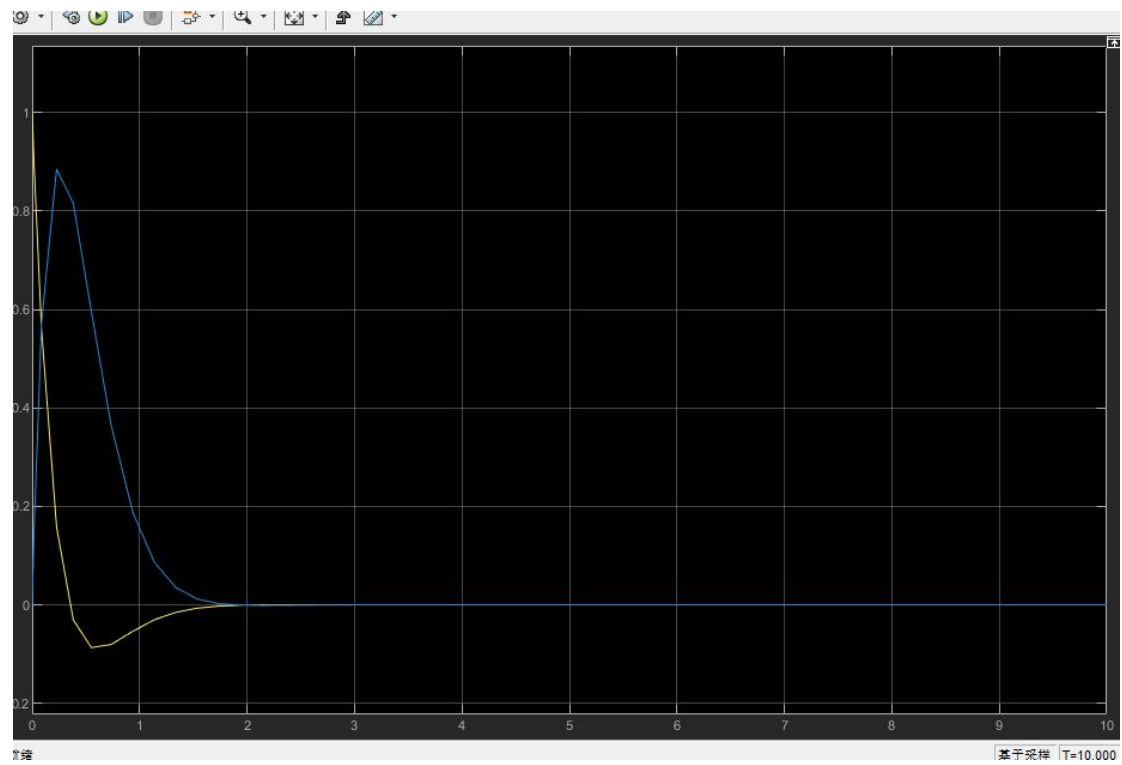
function u = fcn(x)

K1 = [6.158281826222449, 2.051357502843426];
u = -K1 * x;
end

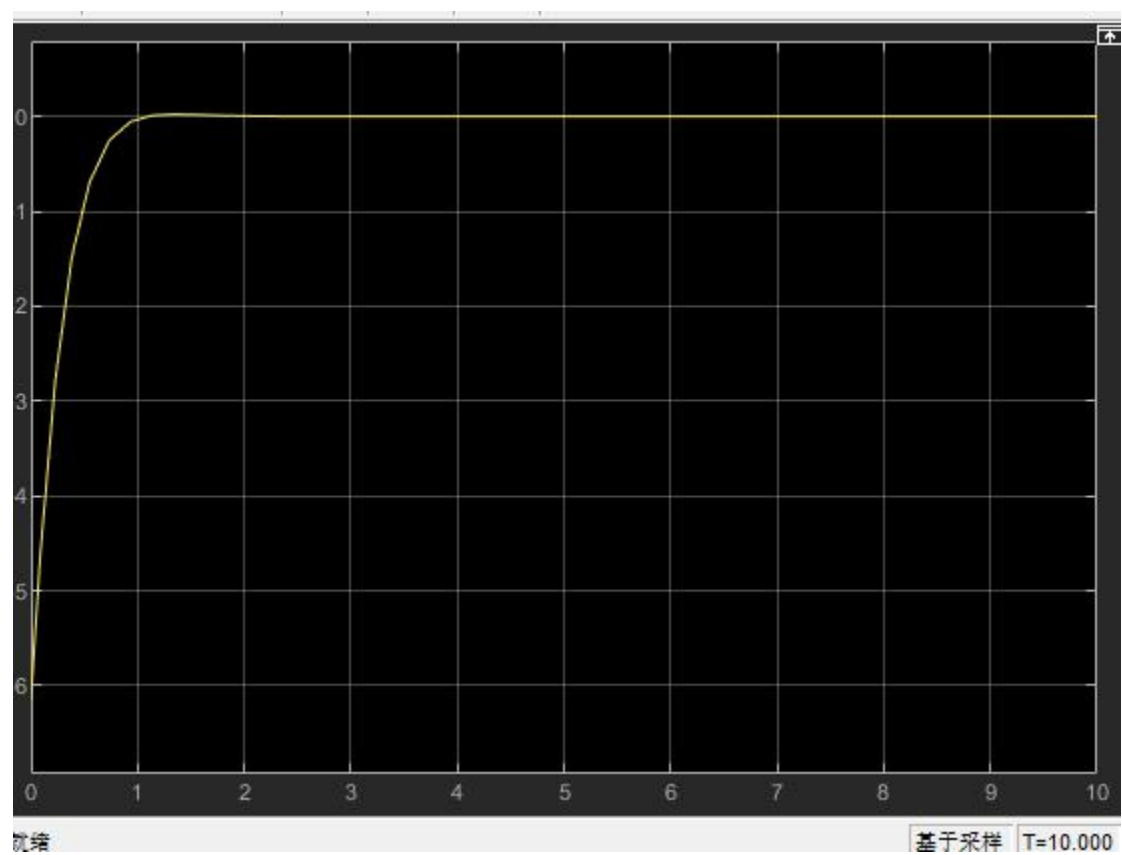
```

Simulink Results:

X1 and x2



Input



Conclusion:

This project used the LQR method to design an optimal controller for the given system. MATLAB and Simulink simulations showed that the controller successfully stabilized the system and reduced the state variables to zero. The results demonstrate that the LQR approach is effective and easy to implement for optimal control problems.